

Rotational Motion

Single Correct Type Questions

1. Match List-I with List-II [28 June, 2022 (Shift-I)]

List-I		List - II	
A.	Moment of inertia of solid sphere of radius R about any tangent.	I.	$\frac{5}{3}MR^2$
B.	Moment of inertia of hollow sphere of radius (R) about any tangent	II.	$\frac{7}{5}MR^2$
C.	Moment of inertia of circular ring of radius (R) about its diameter.	III.	$\frac{1}{4}MR^2$
D.	Moment of inertia of circular disc of radius (R) about any diameter.	IV.	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below:

- (a) A-II, B-I, C-IV, D-III (b) A-I, B-II, C-IV, D-III
(c) A-II, B-I, C-III, D-IV (d) A-I, B-II, C-III, D-IV

2.

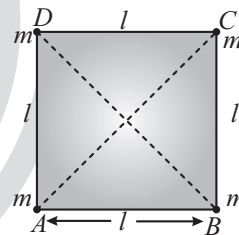
List-I		List - II	
A.	MI of the rod (length L , Mass M , about an axis $\perp$ to the rod Passing through the midpoint)	I.	$8 ML^2/3$
B.	MI of the rod (length L , Mass $2M$, about an axis $\perp$ to the rod Passing through one of its end)	II.	$ML^2/12$
C.	MI of the rod (length $2L$, Mass M , about an axis $\perp$ to the rod Passing through its midpoint)	III.	$ML^2/12$
D.	MI of the rod (length $2L$, Mass $2M$, about an axis $\perp$ to the rod Passing through one of its end)	IV.	$2ML^2/3$

Choose the correct answer from the options given below:

[27 July, 2021 (Shift-I)]

- (a) $A \rightarrow \text{II}, B \rightarrow \text{I}, C \rightarrow \text{III}, D \rightarrow \text{IV}$
(b) $A \rightarrow \text{III}, B \rightarrow \text{IV}, C \rightarrow \text{II}, D \rightarrow \text{I}$
(c) $A \rightarrow \text{III}, B \rightarrow \text{IV}, C \rightarrow \text{II}, D \rightarrow \text{I}$
(d) $A \rightarrow \text{II}, B \rightarrow \text{III}, C \rightarrow \text{I}, D \rightarrow \text{IV}$

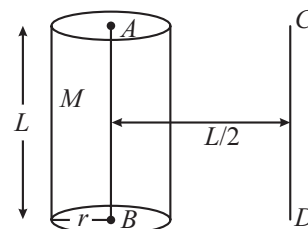
3. Four equal masses, m each are placed at the corners of a square of length (l) as shown in the figure. The moment of inertia of the system about an axis passing through A and parallel to DB would be: [16 March, 2021 (Shift-I)]



- (a) $2ml^2$ (b) ml^2
(c) $\sqrt{3}ml^2$ (d) $3ml^2$

4. The solid cylinder of length 80 cm and mass M has a radius of 20 cm. Calculate the density of the material used if the moment of inertia of the cylinder about an axis CD parallel to AB as shown in figure is 2.7 kg m^2 .

[26 Aug, 2021 (Shift-II)]



- (a) $7.5 \times 10^1 \text{ kg/m}^3$ (b) $1.49 \times 10^2 \text{ kg/m}^3$
(c) 14.9 kg/m^3 (d) $7.5 \times 10^2 \text{ kg/m}^3$

5. Moment of inertia (M.I.) of four bodies, having same mass and radius, are reported as: [24 Feb, 2021 (Shift-I)]

I_1 = M.I. of thin circular ring about its diameter

I_2 = M.I. of circular disc about an axis perpendicular to disc and going through the centre,

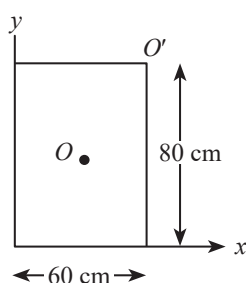
I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then

- (a) $I_1 + I_3 < I_2 + I_4$ (b) $I_1 = I_2 = I_3 < I_4$
 (c) $I_1 = I_2 = I_3 > I_4$ (d) $I_1 + I_2 = I_3 + \frac{5}{2}I_4$

6. [4 Sep, 2020 (Shift-II)]

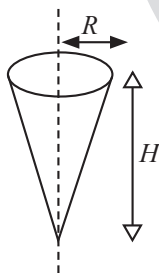


For a uniform rectangular sheet shown in the figure, the ratio of moments of inertia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is:

- (a) $1/2$ (b) $2/3$ (c) $1/4$ (d) $1/8$

7. Shown in the figure is a hollow icecream cone (it is open at the top). If its mass is M , radius of its top, R and height, H , then its moment of inertia about its axis is:

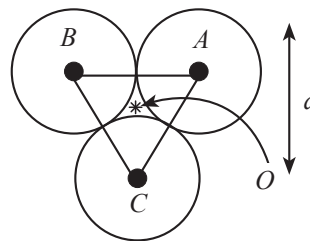
[6 Sep, 2020 (Shift-I)]



- (a) $\frac{MR^2}{2}$ (b) $\frac{M(R^2 + H^2)}{4}$
 (c) $\frac{MH^2}{3}$ (d) $\frac{MR^2}{3}$

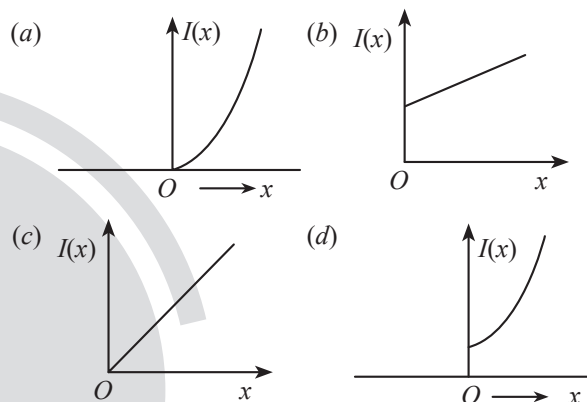
8. Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d . The ratio I_O/I_A of moment of inertia I_O of the system about an axis passing the centroid and about center of any of the spheres I_A and perpendicular to the plane of the triangle is:

[9 Jan, 2020 (Shift-I)]



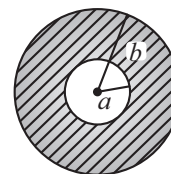
- (a) $\frac{15}{13}$ (b) $\frac{13}{15}$ (c) $\frac{23}{13}$ (d) $\frac{13}{23}$

9. The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is ' $I(x)$ '. Which one of the graphs represents the variation of $I(x)$ with x correctly? [12 Jan, 2019 (Shift-II)]



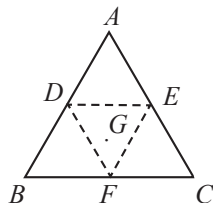
10. A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as $\rho = \rho_0/r$ then the radius of gyration of the disc about its axis passing through the centre is:

[12 April, 2019 (Shift-I)]



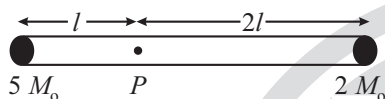
- (a) $\frac{a+b}{2}$ (b) $\frac{a+b}{3}$
 (c) $\sqrt{\frac{a^2 + b^2 + ab}{2}}$ (d) $\sqrt{\frac{a^2 + b^2 + ab}{3}}$

11. The equilateral triangle ABC is cut from a thin solid sheet of wood. (See figure) D , E and F are the mid points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0 . If the smaller triangle DEF is removed from ABC , the moment of inertia of the remaining figure about the same axis is I . Then: [11 Jan, 2019 (Shift-I)]



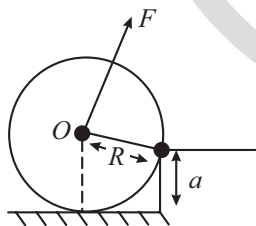
- (a) $I = \frac{15}{16} I_0$ (b) $I = \frac{3}{4} I_0$
 (c) $I = \frac{9}{16} I_0$ (d) $I = \frac{I_0}{4}$

12. A rigid massless rod of length $3l$ has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be : [10 Jan, 2019 (Shift-II)]



- (a) $\frac{g}{13l}$ (b) $\frac{g}{3l}$
 (c) $\frac{g}{2l}$ (d) $\frac{7g}{3l}$

13. A uniform cylinder of mass M and radius R is to be pulled over a step of height a ($a < R$) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is: [2 Sep, 2020 (Shift-I)]



- (a) $Mg\sqrt{1 - \frac{a^2}{R^2}}$ (b) $Mg\sqrt{1 - \left(\frac{R-a}{R}\right)^2}$
 (c) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2 - 1}$ (d) $Mg\frac{a}{R}$

14. A solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is: [26 June, 2022 (Shift-II)]

- (a) $2/5$ (b) $2/7$
 (c) $1/5$ (d) $7/10$

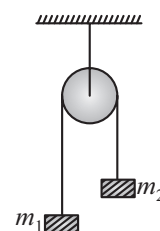
15. Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by:

[27 Aug, 2021 (Shift-II)]

- (a) $\frac{(I_1 - I_2)^2 \omega_1 \omega_2}{2(I_1 + I_2)}$ (b) $\frac{I_1 I_2}{(I_1 + I_2)} (\omega_1 - \omega_2)^2$
 (c) $\frac{I_1 I_2}{2(I_1 + I_2)} (\omega_1 - \omega_2)^2$ (d) $\frac{(\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$

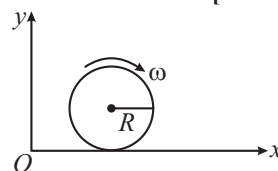
16. A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see figure). A massless string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descends by a distance h is: [9 Jan, 2020 (Shift-II)]

- (a) $\left[\frac{m_1 + m_2}{(m_1 + m_2)R^2 + I} \right]^{\frac{1}{2}} gh$
 (b) $\left[\frac{2(m_1 - m_2)gh}{(m_1 + m_2)R^2 + I} \right]^{\frac{1}{2}}$
 (c) $\left[\frac{2(m_1 + m_2)gh}{(m_1 + m_2)R^2 + I} \right]^{\frac{1}{2}}$
 (d) $\left[\frac{(m_1 + m_2)}{(m_1 + m_2)R^2 + I} \right]^{\frac{1}{2}} gh$



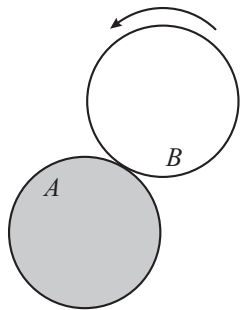
17. A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3} R^2 \omega$. The value of a will be:

[29 June, 2022 (Shift-I)]

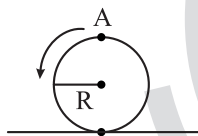


- (a) 2 (b) 3
 (c) 5 (d) 4

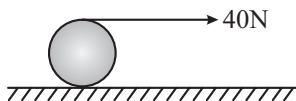
18. A flat surface of a thin uniform disk A of radius R is glued to a horizontal table. Another thin uniform disk B of mass M and with the same radius R rolls without slipping on the circumference of A , as shown in the figure. A flat surface of B also lies on the plane of the table. The center of mass of B has fixed angular speed ω about the vertical axis passing through the center of A . The angular momentum of B is $nM\omega R^2$ with respect to the center of A . Which of the following is the value of n ? [JEE Adv, 2022]



- (a) 2 (b) 5
(c) $7/2$ (d) $9/2$
19. A disc is rolling without slipping on a surface. The radius of the disc is R . At $t = 0$, the top most point on the disc is A as shown in figure. When the disc completes half of its rotation, the displacement of point A from its initial position is [13 April, 2023 (Shift-I)]



- (a) $R\sqrt{(\pi^2 + 4)}$ (b) $R\sqrt{(\pi^2 + 1)}$
(c) $2R$ (d) $2R\sqrt{(1 + 4\pi^2)}$
20. A string is wound around a hollow cylinder of mass 5 kg and radius 0.5m. If the string is now pulled with a horizontal force of 40 N. and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string) [11 Jan, 2019 (Shift-II)]



- (a) 20 rad/s^2 (b) 16 rad/s^2
(c) 12 rad/s^2 (d) 10 rad/s^2
21. A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is : [10 Jan, 2019 (Shift-I)]

- (a) $\frac{3F}{2mR}$ (b) $\frac{F}{3mR}$ (c) $\frac{F}{2mR}$ (d) $\frac{2F}{3mR}$

Integer Type Questions

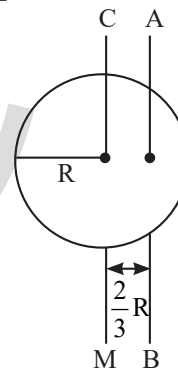
22. The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____ (Assuming the acceleration to be uniform).

[17 March, 2021 (Shift-I)]

23. The moment of inertia of semicircular ring about an axis, passing through the center and perpendicular to the plane of ring, is $\frac{1}{x}MR^2$, where R is the radius and M is the mass of semicircular ring. The value of x will be

[8 April, 2023 (Shift-I)]

24. I_{CM} is the moment of inertia of a circular disc about an axis (CM) passing through its center and perpendicular to the plane of the disc. I_{AB} is its moment of inertia about an axis AB perpendicular to plane and parallel to the axis CM at a distance $\frac{2}{3}R$ from the center. Where R is the radius of the disc. The ratio of I_{AB} and I_{CM} is $x : 9$. The value of x is _____ [25 Jan, 2023 (Shift-I)]



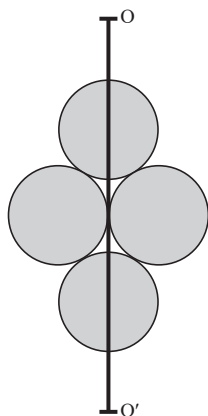
25. If a solid sphere of mass 5 kg and a disc of mass 4 kg have the same radius. Then the ratio of moment of inertia of the disc about a tangent in its plane to the moment of inertia of the sphere about its tangent will be $\frac{x}{7}$. The value of x is _____. [25 Jan, 2023 (Shift-II)]

26. The momentum of inertia of a uniform thin rod about a perpendicular axis passing through on end is I_1 . The same rod is into a ring and its moment of inertia about a diameter is I_2 . If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be _____.

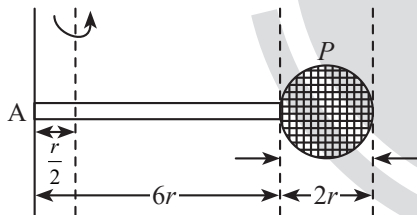
[29 June, 2022 (Shift-II)]

27. Four identical discs each of mass ' M ' and diameter ' a ' are arranged in a small plane as shown in figure. If the moment of inertia of the system about 'OO' is $\frac{x}{4}Ma^2$.

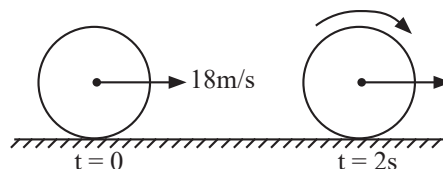
Then, the value of x will be [28 July, 2022 (Shift-I)]



28. Consider a badminton racket with length scales as shown in the figure. If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at, $\frac{r}{2}$ distance from the end A of the handle will _____ Mr^2 . [26 Aug, 2021 (Shift-I)]

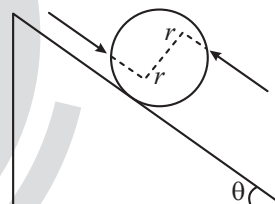


29. A uniform disc of mass 0.5 kg and radius r is projected with velocity 18 m/s at $t = 0$ s on a rough horizontal surface. It starts off with a purely sliding motion at $t = 0$ s. After 2 s it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after 2 s will be _____ J (given, coefficient of friction is 0.3 and $g = 10 \text{ m/s}^2$) [30 Jan, 2023 (Shift-II)]



30. A solid sphere of mass 1 kg and radius 1 m rolls without slipping on a fixed inclined plane with an angle of inclination $\theta = 30^\circ$ from the horizontal. Two forces of magnitude 1 N each, parallel to the incline, act on the sphere, both at distance $r = 0.5$ m from the center of the sphere, as shown in the figure. The acceleration of the sphere down the plane is _____ ms^{-2} . (Take $g = 10 \text{ ms}^{-2}$.)

[JEE Adv, 2022]



ANSWER KEY

- | | | | | | | | | | |
|---------|-----------|---------|----------|---------|---------|---------|----------|----------|------------|
| 1. (a) | 2. (c) | 3. (d) | 4. (b) | 5. (c) | 6. (c) | 7. (a) | 8. (d) | 9. (d) | 10. (d) |
| 11. (a) | 12. (a) | 13. (b) | 14. (b) | 15. (c) | 16. (b) | 17. (c) | 18. (b) | 19. (a) | 20. (b) |
| 21. (d) | 22. [728] | 23. [1] | 24. [17] | 25. [5] | 26. [8] | 27. [3] | 28. [52] | 29. [54] | 30. [2.86] |

EXPLANATIONS

1. (a) Moment of inertia of solid sphere about any tangent,

$$I = I_{cm} + mr^2 = \frac{2}{5}mr^2 + mr^2 = \frac{7}{5}mr^2$$

Moment of inertia of hollow sphere about any tangent,

$$I = I_{cm} + mr^2 = \frac{2}{3}mr^2 + mr^2 = \frac{5}{3}mr^2$$

M.O.I of circular ring about its diameter, $I = \frac{MR^2}{2}$

M.O.I of circular disc about any diameter, $I = \frac{MR^2}{4}$

2. (c) (a) $I = \frac{ML^2}{12}$

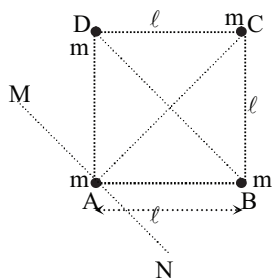
(b) $I = \frac{(2M)L^2}{3}$

(c) $I = \frac{M(2L)^2}{12} = \frac{ML^2}{3}$

(d) $I = \frac{(2M)(2L)^2}{3} = \frac{8ML^2}{3}$

3. (d) $I_{MN} = m(0)^2 + m(\sqrt{2}\ell)^2 + m\left(\frac{\ell}{\sqrt{2}}\right)^2 \times 2$

$$\Rightarrow I_{MN} = 2m\ell^2 + m\ell^2 = 3m\ell^2$$



4. (b) Given, that,

Radius of cylinder, $r = 0.2$ m,
length of cylinder, $L = 0.8$ m
and moment of inertia along CD , $I = 2.7$ kg m².

$$I_{CD} = \frac{1}{2}Mr^2 + M\left(\frac{L}{2}\right)^2$$

$$\Rightarrow 2.7 = M\left(\frac{0.2^2}{2} + \frac{0.8^2}{4}\right)$$

$$\Rightarrow M = 15 \text{ kg}$$

$$\text{Density, } \rho = \frac{M}{\pi r^2 L} = \frac{15}{3.14 \times 0.2^2 \times 0.8} = 1.49 \times 10^2 \text{ kg/m}^3$$

5. (c) We know that,

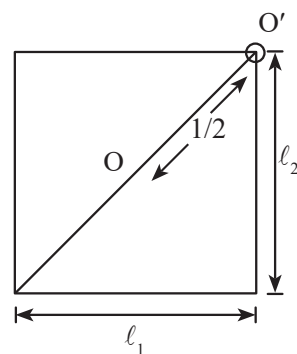
$$I_1 = \frac{1}{2}mr^2, I_2 = \frac{1}{2}mr^2, I_3 = \frac{1}{2}mr^2, I_4 = \frac{2}{5}mr^2$$

$$\therefore I_1 = I_2 = I_3 > I_4$$

6. (c) $\ell_1 = 0.6$ m

$$\ell_2 = 0.8 \text{ m}$$

$$\sqrt{\ell_1^2 + \ell_2^2} = 1$$



Moment of inertia about

$$O, I_1 = \frac{M}{12}(\ell_1^2 + \ell_2^2) \Rightarrow I_1 = \frac{M}{12}$$

Moment of inertia about O' ,

$$I_2 = \frac{M}{12}(\ell_1^2 + \ell_2^2) + \frac{M(\ell_1^2 + \ell_2^2)}{4} = \frac{M}{3}$$

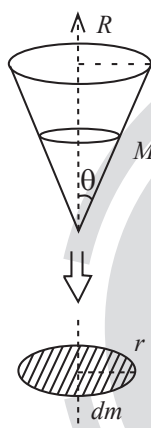
The required ratio, $\frac{I_1}{I_2} = \frac{1}{4}$

7. (a) It can be assumed as stacks of rings having different radii, so $dI = dm r^2$

$$\Rightarrow dm = \left(\frac{M}{\pi R L} \right) 2\pi r dr$$

$$\therefore I = \int_0^H \tan^3 \theta \left(\frac{2h}{RH} \right) x^3 dx$$

$$= \frac{MR^2}{2}$$



8. (d) Moment of inertia, about O ,

$$I_O = 3 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M \left(\frac{d}{\sqrt{3}} \right)^2 \right] = \frac{13}{10} M d^2$$

Moment of inertia about A

$$I_A = 2 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M \left(d^2 \right) \right] + \frac{2}{5} M \left(\frac{d}{2} \right)^2 = \frac{23}{10} M d^2$$

The required ratio, $\frac{I_O}{I_A} = \frac{13}{23}$

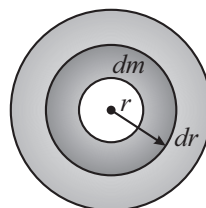
9. (d) $I = I_C + mx^2$

$$I_C = \frac{2}{5} m R^2$$

$$I = \frac{2}{5} m R^2 + mx^2$$

Graph will be parabola opening upward.

10. (d)



$$dI = (dm)r^2 = (\sigma dA)r^2$$

$$= \left(\frac{\sigma_0}{r} 2\pi r dr \right) r^2$$

$$= (\sigma_0 2\pi) r^2 dr$$

$$= I = \int dI = \int_a^b \sigma_0 2\pi r^2 dr$$

$$\sigma_0 2\pi \left(\frac{b^3 - a^3}{3} \right)$$

$$m = \int dm = \int \sigma dA = \sigma_0 2\pi \int_a^b dr$$

$$m = \sigma_0 2\pi (b - a)$$

Radius of gyration

$$k = \sqrt{\frac{I}{m}} = \sqrt{\frac{(b^3 - a^3)}{3(b - a)}}$$

$$k = \sqrt{\frac{(a^2 + b^2 + ab)}{3}}$$

11. (a) Let M is mass and a is side of triangle ABC .

$$\frac{I_{DEF}}{I_0} = \frac{\left(\frac{M}{4} \right) \left(\frac{a}{2} \right)^2}{M(a)^2}$$

$$I_{DEF} = \frac{I_0}{16}$$

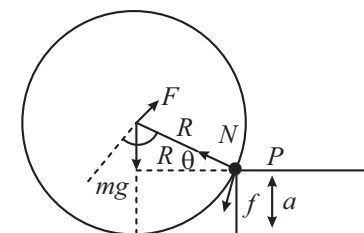
Moment of inertia of remaining sheet $I_0 = \frac{I_0}{16} = \frac{15I_0}{16}$

12. (a) Using torque equation about point P , $\tau = \alpha I$

$$\therefore \alpha = \frac{\tau}{I} = \frac{5M_o g \ell - 4M_o g \ell}{5M_o \ell^2 + 2M_o 4\ell^2}$$

$$= \frac{M_o g \ell}{13M_o \ell^2} = \frac{g}{13\ell}$$

13. (b)



$$FR > mg \cos \theta R$$

$$F > mg \cos \theta$$

$$F > mg \sqrt{\frac{R^2 - (R-a)^2}{R}} \Rightarrow Mg \sqrt{1 - \left(\frac{R-a}{R}\right)^2}$$

$$14. (b) K_R = \frac{1}{2} I \omega^2 = \frac{1}{2} \times \frac{2}{5} MR^2 \times \frac{v^2}{R^2} = \frac{Mv^2}{5}$$

$$K_T = \frac{1}{2} I \omega^2 + \frac{1}{2} Mv^2 = \frac{Mv^2}{5} + \frac{1}{2} Mv^2 = \frac{7Mv^2}{10}$$

$$\text{Thus, } \frac{K_R}{K_T} = \frac{2}{7}$$

15. (c) From conservation of Angular momentum,

Initial angular momentum = Final angular momentum

$$\Rightarrow I_1 \omega_1 + I_2 \omega_2 = (I_1 + I_2) \omega$$

$$\Rightarrow \omega = \frac{I_1 \omega_1 + I_2 \omega_2}{I_1 + I_2} \quad \dots(i)$$

Initial kinetic energy,

$$KE_i = \frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 \quad \dots(ii)$$

Final kinetic energy,

$$KE_f = \frac{1}{2} (I_1 + I_2) \omega^2 \quad \dots(iii)$$

From equation (i), (ii) and (iii),

$$\Rightarrow |KE_f - KE_i| = \frac{1}{2} \left(\frac{I_1 I_2}{I_1 + I_2} \right) (\omega_1 - \omega_2)^2$$

$$16. (b) K_i + U_i = K_f + U_f$$

$$0 + 0 = \frac{1}{2} m_2 v^2 + \frac{1}{2} m_1 v^2 + \frac{1}{2} I \omega^2 - m_1 gh + m_2 gh$$

$$(m_1 - m_2) gh = \frac{1}{2} m_2 (\omega R)^2 + \frac{1}{2} m_1 (\omega R)^2 + \frac{1}{2} I \omega^2$$

$$\sqrt{\frac{2(m_1 - m_2) gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right) R^2}} = \omega$$

$$17. (c) L = I_{cm} \omega + Mv_{cm} R$$

$$= \left(\frac{2}{3} R^2 \right) \omega + (1)(R\omega)R = \frac{5}{3} R^2 \omega$$

$$\Rightarrow a = 5$$

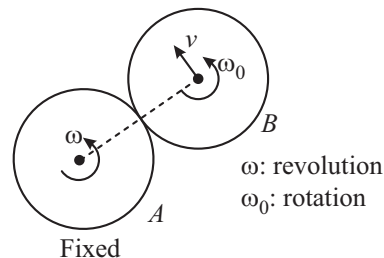
18. (b) If the linear velocity of center of mass of disc B is v , then

$$v = \omega (2R)$$

Further, since the disc B is rolling without slipping,

$$v = \omega_0 R : \text{no slipping}$$

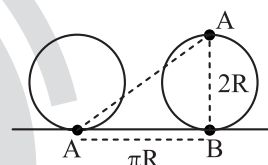
$$\therefore \omega_0 = 2\omega$$



$$\vec{L} = m\vec{r} \times \vec{v}_c + I_c \omega_0 = M^2 R v + \frac{1}{2} MR^2 \omega_0$$

$$= 4MR^2 \omega + \frac{1}{2} MR^2 (2\omega) = 5MR^2 \omega \quad \therefore n = 5$$

19. (a)

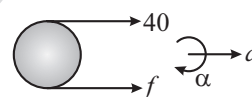


$$\text{Displacement} = \sqrt{(2R)^2 + (\pi R)^2} = R\sqrt{\pi^2 + 4}$$

$$20. (b) 40 + f = m(R\alpha) \quad \dots(i)$$

$$40 \times R - f \times R = mR^2 \alpha$$

$$40 - f = mR\alpha \quad \dots(ii)$$



From (i) and (ii)

$$\alpha = \frac{40}{mR} = 16 \text{ rad/s}^2$$

$$21. (d) F - f = ma$$

$$fR = \frac{1}{2} mR^2 \alpha$$

$$a = \alpha R$$

From above equations,

$$F = \frac{3}{2} mR\alpha$$

$$\text{and } \alpha = \frac{2F}{3mR}$$

$$22. [728] \quad \omega_f = 2460 \times \frac{2\pi}{60} = 82\pi$$

$$\omega_i = \frac{900 \times 2\pi}{60} = 30\pi$$

$$\alpha = \frac{\omega_f - \omega_i}{t} = \frac{82\pi - 30\pi}{26}$$

$$= 2\pi \text{ rad/sec}^2$$

$$\theta = \frac{\omega_f^2 - \omega_i^2}{2\alpha} = \frac{(82\pi + 30\pi)(82\pi - 30\pi)}{2 \times 2\pi}$$

$$= \frac{(112 \times 52)\pi^2}{4\pi} = 2\pi n$$

$$\text{No. of revolution } (n) = \frac{(112 \times 52)\pi}{2\pi} = 728$$

23. [1] Moment of inertia of full ring of mass (2m) and radius (R) is

$$I_{\text{Ring}} = 2MR^2$$

Semicircular ring has same radius (R) and mass is half (M.)

So, moment of inertia of Semicircular ring,

$$I_{\text{semicircular ring}} = \frac{I_{\text{Ring}}}{2} = \frac{2MR^2}{2} = MR^2$$

$$\therefore x = 1$$

24. [17] Moment of inertia of a circular disc about centre of mass and perpendicular to its plane $I_{cm} = \frac{mR^2}{2}$

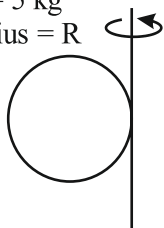
Using parallel axis theorem

$$I_{AB} = \frac{mR^2}{2} + m\left(\frac{2R}{3}\right)^2 = \frac{17}{18}mR^2$$

$$\therefore \frac{I_{AB}}{I_{cm}} = \frac{17}{9} \Rightarrow x = 17$$

25. [5] $m_1 = 5 \text{ kg}$
Radius = R

Solid sphere



Moment of inertia of a solid sphere about an axis passing through its center,

$$I_1 = \frac{2}{5}m_1R^2$$

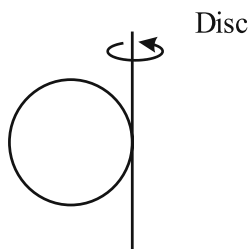
Using theorem of parallel axis,

$$I_1 = \frac{2}{5}m_1R^2 + m_1R^2 \quad I_1 = m_1R^2\left(\frac{7}{5}\right)$$

$$\Rightarrow I_1 = 7R^2$$

$$[\because m_1 = 5\text{kg}]$$

For a disc, moment of inertia about an axis passing along its diameter $I = \frac{m_2R^2}{4}$



Using theorem of parallel axis,

$$I_2 = \frac{m_2R^2}{4} + m_2R^2$$

$$I_2 = \frac{5}{4}m_2R^2$$

$$I_2 = 5R^2$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{5}{7}$$

$$x = 5$$

$$26. [8] \quad I_1 = \frac{ML^2}{3}$$

$$\text{Now, } 2\pi r = L$$

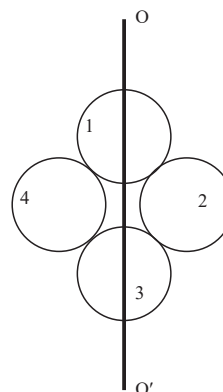
$$\Rightarrow r = \frac{L}{2\pi}$$

$$I_2 = \frac{1}{2}Mr^2 = \frac{1}{2} \frac{ML^2}{4\pi^2}$$

$$\text{So, } \frac{I_1}{I_2} = \frac{8\pi^2}{3} \Rightarrow x = 8$$

$$27. [3] \quad I_1 = I_3 = \frac{Ma^2}{4 \times 4}, I_2 = I_4 = \frac{Ma^2}{4 \times 4} + \frac{Ma^2}{4}$$

$$I_{00'} = I_1 + I_2 + I_3 + I_4$$



$$I_{oo'} = \frac{6Ma^2}{8} = \frac{3}{4}Ma^2$$

$$\Rightarrow x = 3$$

$$\begin{aligned} 28. [52] \quad I_{xx'} &= \left[I_{y'} + M \left(\frac{5r}{2} \right)^2 \right] + \left[I_{zz'} + M \left(\frac{13r}{2} \right)^2 \right] \\ &= M \left(\frac{36r^2}{12} \right) + \frac{M \times 25r^2}{4} + \frac{Mr^2}{2} + M \times \frac{169Mr^2}{4} \\ &= 52Mr^2 \end{aligned}$$

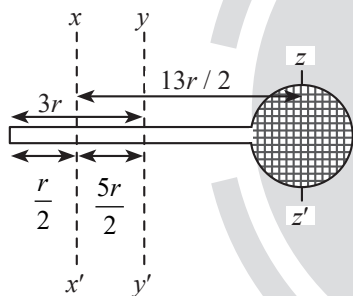
$$29. [54] \quad a = -\mu_k g = -3$$

$$V = 18 - 3 \times 2$$

$$V = 12 \text{ m/s}$$

$$KE = \frac{1}{2}mv^2 + \frac{1}{2} \frac{mr^2}{2} \frac{v^2}{r^2} = \frac{3}{4}mv^2$$

$$KE = 3 \times 18 = 54 \text{ J}$$



Assuming circular portion to be disc

30.[2.86] Translation motion of COM:

$$5 + 1 - 1 - f = 1a$$

$$5 - f = a$$

...(i)

Rotational motion about COM:

$$f \cdot 1 - 2(1(0.5)) = \frac{2}{5}Mr^2\alpha$$

$$\Rightarrow f - 1 = \frac{2}{5}a \Rightarrow f = 1 + \frac{2}{5}a$$

...(ii)

From equations (i) and (ii),

$$5 - a = 1 + \frac{2}{5}a$$